

Look at the 250% leverage ratio case. Now, for the same doubling-the-market result, at a 5% margin, the numbers increase to 53.8%. This is a 53.8% chance (better than one in two) that you will earn more than a 30% return on your original investment. In fact, the higher your leverage ratio, the better your odds up to a point. It's not surprising that so many have rushed to leverage their capital: The odds look so good.

Now, however, let's examine "the bad," the downside. What are the odds that you will lose money? What are the odds that, with leverage, you will end up with less than when you started?

Table 4.3 shows what happens when things go wrong. Although the odds of doubling the market are much greater for the leverage case, the odds of losing money are higher as well. For example, if you leverage yourself at 100% with a margin cost of 5%, the odds of losing money—that is, the chances of having less than your original equity capital at year's end—are 33.8%.

TABLE 4.3 Leverage: The Bad—What Are the Chances of Losing Money? (Market Expected Return: 15%; Volatility: 30%)

Leverage Ratio	Interest Cost		
	3%	5%	7%
0%	30.9%	30.9%	30.9%
25%	31.6%	32.0%	32.5%
50%	32.0%	32.8%	33.6%
100%	32.6%	33.8%	35.1%
250%	33.4%	35.2%	36.9%

Source: Moshe Milevsky and the IFID Centre, 2008.

Thus, if you start with \$10,000 and borrow \$10,000, at year's end, after paying back your loan with interest, the probability of having less than \$10,000 is 33.8%.

"Wait a minute," you might say. As you can see, 33.8% is not that much higher than the 30.9% chance of losing money if you don't use leverage at all. In other words, there's only a 3% greater chance that using leverage will court disaster. That doesn't sound so bad, does it? Is that the sum total of your risk? Well, not exactly. Here's "the ugly."